1 Waves and Diffusions

1.1 The Wave Equation

Definition 1.1.1 (One Dimensional Wave Equation on $\mathbb{R}$):

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$

□

The wave equation is among the simplest second-order equation since you can easily convert it to a first-order equation by:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = \left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right) u = 0$$

□

Theorem 1.1.1 (General Solution of the One Dimensional Wave Equation): Consider the equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$

Then, the general solution is:

$$u(x,t) = f(x+ct) + g(x-ct)$$

where, f, g are twice differential functions.

Proof,

Recall that we can factor the equation as:

$$\left(\frac{\partial}{\partial t}+c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t}-c\frac{\partial}{\partial x}\right)u=0$$

We have two methods to solve it.

(Method 1)

In this method, we convert the second order of PDE into a system of first order PDE. We do so by using the substitution:

$$v=\frac{\partial u}{\partial t}-c\frac{\partial u}{\partial x}$$

Thus, we have:

$$\frac{\partial v}{\partial t}+c\frac{\partial v}{\partial x}=0$$

Now, by method of characteristic, this implies that solution is constant along the line having the derivative:

$$\frac{\mathrm{d}x}{\mathrm{d}t}=c$$

Of course, the equation of this line is:

$$x=ct+C\Longrightarrow x-ct=C$$

Thus,

$$v(x,t)=h(x-ct)$$

Recalling the definition,

$$\frac{\partial u}{\partial t}-c\frac{\partial u}{\partial x}=h(x-ct)$$

The solution for u is thus then the superposition of the solution of the homogenous system as well as the particular solution for the equation above.

For the homogenous system, we need to solve:

$$\frac{\partial u}{\partial t} - c\frac{\partial u}{\partial x} = 0$$

Similarly to how we solved for v, this gives us:

$$u(x,t) = f(x+ct)$$

For the particular solution, we guess that:

$$u(x,t) = g(x-ct)$$

Now,

$$\frac{\partial u}{\partial t} - c\frac{\partial u}{\partial x} = -cg'(x-ct) - cg'(x-ct) = -2cg'(x-ct)$$

Thus,

$$g(s) = -\frac{1}{2c}\int \mathrm{d}s\, h(s)$$

Hence,

$$u(x,t) = f(x+ct) + g(x-ct)$$

The reason why the solution is a homogenous + particular has to do with linearity.

(Method 2)

In this method, we use method of characteristic to do a change of variable.

From the factorization, it's clear that solutions can be either constant along:

$$\xi = x + ct \quad \eta = x - ct$$

We now solve $u(x,t)$ by doing a change of coordinates to (ξ, η).

1.2 Diffusion on the Whole Line

1.2.1 Exercises

Exercise 1.2.1.1. Prove from first principles that $Q(x,t)$ must have the form:

$$Q(x,t) = g\left(\frac{x}{\sqrt{4kt}}\right)$$

(a) Assuming uniqueness show that $Q(x,t) = Q(\sqrt{a}x, at)$. This identity is valid for all $a, t \in \mathbb{R}^+, x \in \mathbb{R}$.

(b) Choose $a = \frac{1}{4kt}$

Solution,

We consider the Dirichlet problem for the diffusion on the real line.

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \quad u(x,0) = \varphi(x) \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$

We wish to find a $Q(x,t)$ which solves the diffusion equation under the particular initial condition:

$$Q(x,0) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases} = \operatorname{sgn}(x)$$

Thanks to uniqueness, scaling Q doesn't change the solution. Of course, "scaling" here is not the usual scaling since $Q(ax, bt)$ might not solve the equation.

Recall that $Q(\sqrt{a}x, at)$ however still solves the problem. (It's a property we've proven in the exercises)

Therefore, by uniqueness,

$$Q(x,t) = Q\big(\sqrt{a}x, at\big)$$

Where, $a \in \mathbb{R}^+$

To see why, this happens, it's because:

$$\operatorname{sgn}(ax) = \operatorname{sgn}(x)$$

if $a \in \mathbb{R}^+$.

Now, if we pick $a = \frac{1}{4kt}$ it must necessarily follow that:

$$Q(x,t) = Q\left(\frac{x}{\sqrt{4kt}}, \frac{t}{4kt}\right)$$

Note, that this is valid because $t \in \mathbb{R}^+$. Now, finally, we have:

$$Q(x,t) = Q\left(\frac{x}{\sqrt{4kt}}, \frac{1}{4k}\right)$$

Hence, Q is only dependent on $\frac{x}{\sqrt{4kt}}$. Therefore, it follows:

$$Q(x,t) = g(p) \quad p = \frac{x}{\sqrt{4kt}}$$

□

Exercise 1.2.1.2. Let $\varphi(x)$ be a continuous function such that $|\varphi(x)| \leq Ce^{ax^2}$. Show that formula (8) for the solution of the diffusion equation makes sense for $0 < t < \frac{1}{4ak}$, but not necessarily for larger t

Solution,

Recall (8):

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \mathrm{d}y \, e^{-\frac{(x-y)^2}{4kt}} \varphi(y)$$

Solves the Dirichlet problem for the diffusion equation on the real line. i.e.,

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \quad u(x,0) = \varphi(x) \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$

Now, taking the absolute value of u we get:

$$|u(x,t)| \leq \frac{C}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \mathrm{d}y \, e^{-\frac{(x-y)^2}{4kt} + ax^2}$$

The exponent,

$$-\frac{(x-y)^2}{4kt} + ax^2 = \frac{(1-4akt)x^2 + 2xy - y^2}{4kt}$$

For the critical case $t = \frac{1}{4akt}$, the coefficient of x^2 vanishes. Thus, the coefficient becomes:

$$\frac{y(2x-y)}{4kt}$$

The issue becomes that the solution doesn't immediately decrease thanks to the $(2x-y)$ term.

Of course, it would require the $\varphi(x)$ to be at the edge of its condition. However, it is still nonetheless possible.

Thanks to this non-decreasing nature of the solution, it would follow that the further out you go, the hotter it would become.

This effect is magnified for any t larger.

□

Exercise 1.2.1.3. Prove the uniqueness of the diffusion problem with Neumann boundary conditions:

$$\frac{\partial u}{\partial t} - k\frac{\partial^2 u}{\partial x^2} = f(x,t) \quad (x,t) \in (0,l) \times \mathbb{R}^+ \quad u(x,0) = \varphi(x)$$
$$\frac{\partial u}{\partial x}(0,t) = g(t) \quad \frac{\partial u}{\partial x}(l,t) = h(t)$$

by the energy method.

Proof,

Let u_1, u_2 be two solutions to the Neumann problem of the diffusion equation. Consider,

$$w = u_1 - u_2$$

Now, by superposition, w is also a solution. Now, multiply it by w and we get:

$$\left(\frac{\partial w}{\partial t} - k\frac{\partial^2 w}{\partial x^2}\right) w = 0$$

$$w\frac{\partial w}{\partial t} - kw\frac{\partial^2 w}{\partial x^2} = 0$$

$$\frac{\partial}{\partial t}\left[\frac{1}{2}w^2\right] + \frac{\partial}{\partial x}\left(-k\frac{\partial w}{\partial x}w\right) + k\frac{\partial^2 w}{\partial x^2} = 0$$

Integrating over the interval $x \in (0, l)$, we get:

$$\begin{aligned} 0 &= \int_0^l \mathrm{d}x \, \frac{\partial}{\partial t}\left[\frac{1}{2}w^2\right] + \frac{\partial}{\partial x}\left(-k\frac{\partial w}{\partial x}w\right) + k\frac{\partial^2 w}{\partial x^2} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\int_0^l \mathrm{d}x \, \frac{1}{2}w^2 + \left[-k\frac{\partial w}{\partial x}w\right]_0^l + k\left[\frac{\partial w}{\partial x}\right]_0^l \end{aligned}$$

Simplifying, we get:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_0^l \mathrm{d}x \, \frac{1}{2}w^2\right) - k(0-0) + k(0-0) = 0$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_0^l \mathrm{d}x \, \frac{1}{2}w^2\right) = 0$$

Hence, the quantity:

$$\int_0^l \mathrm{d}x \, w^2$$

is constant over time.

Since,

$$w(x, 0) = 0 \Rightarrow \int_0^l \mathrm{d}x \, w(x, 0)^2 = 0$$

Therefore, it follows that:

$$w(x, t) \equiv 0 \Rightarrow u_1 = u_2$$

Hence, we're done.

□

Exercise 1.2.1.4. Solve the diffusion equation with constant dissipation:

$$\frac{\partial u}{\partial t} - k\frac{\partial^2 u}{\partial x^2} + bu = 0 \quad x \in \mathbb{R}$$
$$u(x,0) = \varphi(x)$$

where, $b \in \mathbb{R}^+$.

Solution,

Let,

$$u(x,t) = e^{-bt}v(x,t)$$

So,

$$\frac{\partial u}{\partial t} = e^{-bt}\left(-bv + \frac{\partial v}{\partial t}\right) \quad \frac{\partial^2 u}{\partial x^2} = e^{-bt}\frac{\partial^2 v}{\partial x^2}$$

Substituting back we get:

$$e^{-bt}\left(-bv + \frac{\partial v}{\partial t} - k\frac{\partial^2 v}{\partial x^2} + bv\right) = 0$$
$$\frac{\partial v}{\partial t} = k\frac{\partial^2 v}{\partial x^2}$$

So, v satisfies the following Dirichlet problem for the diffusion equation:

$$\frac{\partial v}{\partial t} = k\frac{\partial^2 v}{\partial x^2} \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$
$$v(x,0) = e^{b0}\varphi(x) = \varphi(x)$$

Recall the solution of the Dirichlet problem is given by:

$$v(x,t) = \frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty} \mathrm{d}y\, e^{-\frac{(x-y)^2}{4kt}}\varphi(y)$$

Thus,

$$u(x,t) = \frac{e^{-bt}}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \mathrm{d}y \, e^{-\frac{(x-y)^2}{4kt}} \varphi(y)$$

□

Exercise 1.2.1.5. Solve the diffusion equation with the variable dissipation:

$$\frac{\partial u}{\partial t} - k\frac{\partial^2 u}{\partial x^2} + bt^2 u = 0 \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$
$$u(x,0) = \varphi(x)$$

Solution,

Here, we can re-use the integrating factor method from ODE.

We're given the equation:

$$\frac{\partial u}{\partial t} + bt^2 u = k\frac{\partial^2 u}{\partial x^2}$$

By ignoring the diffusion term, we get:

$$\frac{\mathrm{d}u}{\mathrm{d}t} + bt^2 u = 0$$

Recall that by the method of integrating factor, we can convert this into:

$$\frac{\mathrm{d}w}{\mathrm{d}t} = 0$$

where, $w = Iu$ where, I is the integrating factor. The integrating factor is computed as:

$$I = \exp \int \mathrm{d}t \, bt^2 = e^{\frac{bt^3}{3}}$$

This gives us the required change of variable as:

$$u(x,t) = e^{-\frac{bt^3}{3}} v(x,t)$$

Because of I only being dependent on t, it will pass through $\frac{\partial^2}{\partial x^2}$. Hence, using the transformation, we convert the equation to:

$$\frac{\partial v}{\partial t} = kI\frac{\partial^2 u}{\partial x^2} = k\frac{\partial^2}{\partial x^2}(Iu) = k\frac{\partial^2 v}{\partial x^2}$$

The initial condition given is:

$$v(x,0) = I(0)u(x,0) = (1)\varphi(x) = \varphi(x)$$

Hence,

$$v(x,t) = \frac{1}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty} \mathrm{d}y\, e^{-\frac{(x-y)^2}{4kt}}\varphi(y)$$

Therefore,

$$u(x,t) = \frac{e^{-\frac{bt^3}{3}}}{\sqrt{4\pi kt}}\int_{-\infty}^{\infty} \mathrm{d}y\, e^{-\frac{(x-y)^2}{4kt}}\varphi(y)$$

□

Exercise 1.2.1.6. Solve the heat equation with convection:

$$\frac{\partial u}{\partial t} - k\frac{\partial^2 u}{\partial x^2} + V\frac{\partial u}{\partial x} = 0 \quad (x,t) \in \mathbb{R}\times\mathbb{R}^+$$
$$u(x,0) = \varphi(x)$$

where V is constant.

Solution,

Let, $y = x - Vt$, $v(y,t) = u(x,t) = u(y+Vt,t)$

Now, by the chain-rule:

$$\frac{\partial v}{\partial t} = V\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} \quad \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} \quad \frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 u}{\partial x^2}$$

Therefore, we get:

$$\left(\frac{\partial v}{\partial t} - V\frac{\partial u}{\partial y}\right) - k\frac{\partial^2 v}{\partial y^2} + V\frac{\partial v}{\partial y} = 0$$

$$\frac{\partial v}{\partial t} - k\frac{\partial^2 v}{\partial y^2} = 0$$

which is the diffusion equation.

Now, the initial condition changes to:

$$v\big(y,0\big) = u(y + V(0), 0) = u(y, 0) = \varphi(y)$$

Hence, it follows that:

$$v(y,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \mathrm{d}z\, e^{-\frac{(y-z)^2}{4kt}} \varphi(z)$$

Since, using the fact that $y = x - Vt$, we get:

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \mathrm{d}y\, e^{-\frac{(x-Vt-z)^2}{4kt}} \varphi(y)$$

Of course, we also used the fact that z is the dummy variable to change it to y. If this feels weird, just use the convolution formula:

$$v(y,t) = H * \varphi(y) \Longrightarrow u(x,t) = H * \varphi(x - Vt)$$

and so we're done.

□

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2 Reflections and Sources

2.1 Diffusion on the Half-Life

Theorem 2.1.1 (Diffusion on the Half-Line): Consider the diffusion equation on the half life with the Dirichlet boundary condition:

$$\frac{\partial u}{\partial t} - k\frac{\partial^2 u}{\partial x^2} = 0 \quad (x,t) \in \mathbb{R}^+ \times \mathbb{R}^+$$
$$u(x,0) = \varphi(x) \quad u(0,t) = 0$$

has the solution:

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \int_{\mathbb{R}^+} \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right] \varphi(y)$$

"Proof",

For this "proof", we don't dwell much on the matters of analysis.

Recall that we can solve the problem for the whole line. So, what we do is extend the problem.

$$\varphi_{\text{odd}}(x) = \begin{cases} \varphi(x) & x > 0 \\ 0 & x = 0 \\ -\varphi(-x) & x < 0 \end{cases}$$

is called the odd-extension of φ.

Using this initial value, we wish to solve the Dirichlet boundary problem:

$$\frac{\partial u}{\partial t} - k\frac{\partial^2 u}{\partial x^2} = 0 \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$
$$u(x,0) = \varphi_{\text{odd}}(x)$$

We recall that the solution is:

$$u(x,t) = \int_{\mathbb{R}^+} \mathrm{d}y\, S(x-y,t)\varphi_{\text{odd}}(y)$$

where,

$$S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}$$

Consider the restriction of u on $\mathbb{R}^+$. Call it v.

Then, v satisfies the problem:

$$\frac{\partial v}{\partial t} - k\frac{\partial^2 v}{\partial x^2} = 0 \quad (x,t) \in \mathbb{R}^+ \times \mathbb{R}^+$$
$$v(x,0) = \varphi_{\text{odd}}(x) = \varphi(x)$$

Since, S is even for x, it follows that:

$$S(x-y,t)\varphi_{\text{odd}}(t)$$

is odd. Hence,

$$v(0,t) = 0$$

By the uniqueness of solution, it follows that v is the unique solution of the diffusion on the half-line.

Now, we can also rewrite:

$$v(x,t) = \int_{\mathbb{R}} \mathrm{d}y\, S(x-y,t)\varphi_{\text{odd}}(y)$$

in terms of φ. This isn't very difficult, only requiring a bit of algebra and calculus (change of variables). In the end of the computation, we get:

$$v(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{\mathbb{R}^+} \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right] \varphi(y)$$

□

Example: Solve the problem:

$$\frac{\partial v}{\partial t} - k\frac{\partial^2 v}{\partial x^2} = 0 \quad (x,t) \in \mathbb{R}^+ \times \mathbb{R}^+$$
$$v(x,0) = 1 \quad v(0,t) = 0$$

Solution,

Recall that the solution of the general solution is:

$$\begin{aligned} v(x,t) &= \frac{1}{\sqrt{4\pi kt}} \int_{\mathbb{R}^+} \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right] \varphi(y) \\ &= \frac{1}{\sqrt{4\pi kt}} \int_{\mathbb{R}^+} \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right] (1) \\ &= \frac{1}{\sqrt{4\pi kt}} \int_{\mathbb{R}^+} \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right] \end{aligned}$$

Now, using the substitutions:

$$p = \frac{x-y}{\sqrt{4kt}} \quad q = \frac{x+y}{\sqrt{4kt}}$$

we can transform the integrals into:

$$\begin{aligned} v(x,t) &= \frac{1}{\sqrt{4\pi kt}} \left[\int_{\frac{x}{\sqrt{4kt}}}^{-\infty} \mathrm{d}p\, e^{-p^2} \left(-\sqrt{4kt} \right) - \int_{\frac{x}{\sqrt{4kt}}}^{\infty} \mathrm{d}q\, e^{-q^2} \left(\sqrt{4kt} \right) \right] \\ &= \frac{1}{\sqrt{4\pi kt}} \left[\sqrt{4kt} \left(\int_{-\infty}^{\frac{x}{\sqrt{4kt}}} \mathrm{d}p\, e^{-p^2} - \int_{\frac{x}{\sqrt{4kt}}}^{\infty} \mathrm{d}q\, e^{-q^2} \right) \right] \\ &= \frac{1}{\sqrt{\pi}} \left[\int_{-\infty}^{\frac{x}{\sqrt{4kt}}} \mathrm{d}p\, e^{-p^2} - \int_{\frac{x}{\sqrt{4kt}}}^{\infty} \mathrm{d}q\, e^{-q^2} \right] \end{aligned}$$

Recall,

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \mathrm{d}t\, e^{-t^2}$$

So,

$$
\begin{aligned}
v(x,t) &= \frac{1}{\sqrt{\pi}}\left[\sqrt{\pi}\left(1+\frac{1}{2}\operatorname{erf}\left(\frac{x}{\sqrt{4kt}}\right)\right)-\sqrt{\pi}\left(1-\frac{1}{2}\operatorname{erf}\left(\frac{x}{\sqrt{4kt}}\right)\right)\right] \\
&= 1-\frac{1}{2}\operatorname{erf}\left(\frac{x}{\sqrt{4kt}}\right)-1+\frac{1}{2}\operatorname{erf}\left(\frac{x}{\sqrt{4kt}}\right) \\
&= \operatorname{erf}\left(\frac{x}{\sqrt{4kt}}\right)
\end{aligned}
$$

Therefore,

$$
v(x,t) = \operatorname{erf}\left(\frac{x}{\sqrt{4kt}}\right)
$$

□

Theorem 2.1.2 (Diffusion on half-life (Neumann)): Consider the problem:

$$
\frac{\partial w}{\partial t} - k\frac{\partial^2 w}{\partial x^2} = 0 \quad (x,t) \in \mathbb{R}^+ \times \mathbb{R}^+
$$

$$
w(x,0) = \varphi(x) \quad \frac{\partial w}{\partial x}(0,t) = 0
$$

Then, w exists and is unique and is given by:

$$
w(x,t) = \frac{1}{\sqrt{4\pi kt}}\int_{\mathbb{R}^+} \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} + e^{-\frac{(x+y)^2}{4kt}}\right]\varphi(y)
$$

Proof,

Here, we use the even extension:

$$
\varphi_{\text{even}}(x) = \begin{cases} \varphi(x) & x \geq 0 \\ \varphi(-x) & x < 0 \end{cases}
$$

Then, we can consider the problem of diffusion on $\mathbb{R}$. I.e.,

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0 \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+$$
$$u(x,0) = \varphi_{\text{even}}(x)$$

Now, the solution to this problem exists and is unique. Moreover, it is given by:

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \int_{-\infty}^{\infty} \mathrm{d}y \, e^{-\frac{(x-y)^2}{\sqrt{4kt}}} \varphi_{\text{even}}(y)$$

Since we wish to write our answer in terms of $\varphi(y)$, we can rearrange the integral and use symmetry of φ_{even} to get:

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \left[\int_0^{\infty} \mathrm{d}y \, S(x-y,t)\varphi(y) + \int_{-\infty}^{0} \mathrm{d}y \, S(x-y,t)\varphi(y) \right]$$

By reflecting the second integral,

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \left[\int_0^{\infty} \mathrm{d}y \, S(x-y,t)\varphi(y) + \int_{0}^{-\infty} \mathrm{d}y \, S(x+y,t)\varphi(-y) \right]$$

Since, $\varphi(-y) = \varphi(y)$, we can combine the two integrals to get:

$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \int_0^{\infty} \mathrm{d}y \, (S(x-y,t) + S(x+y,t))\varphi(y)$$
$$u(x,t) = \frac{1}{\sqrt{4\pi k t}} \int_0^{\infty} \mathrm{d}y \left(e^{-\frac{(x-y)^2}{\sqrt{4kt}}} + e^{-\frac{(x+y)^2}{\sqrt{4kt}}} \right) \varphi(y)$$

Differentiating wrt x and then under the integral sign:

$$\begin{aligned} \frac{\partial u}{\partial x}(x,t) &= \frac{1}{\sqrt{4\pi k t}} \int_0^{\infty} \mathrm{d}y \, \frac{\partial}{\partial x} \left[\left(e^{-\frac{(x-y)^2}{\sqrt{4kt}}} + e^{-\frac{(x+y)^2}{\sqrt{4kt}}} \right) \varphi(y) \right] \\ &= \frac{1}{4kt\sqrt{\pi}} \int_0^{\infty} \mathrm{d}y \left(e^{-\frac{(x-y)^2}{\sqrt{4kt}}} (x-y) + e^{-\frac{(x+y)^2}{\sqrt{4kt}}} (x+y) \right) \varphi(y) \end{aligned}$$

So, setting $x = 0$, we get:

$$\frac{\partial u}{\partial x}(0,t) = \frac{1}{4kt\sqrt{\pi}} \int_0^\infty \mathrm{d}y \left(e^{-\frac{(-y)^2}{\sqrt{4kt}}}(-y) + e^{-\frac{(y)^2}{\sqrt{4kt}}}(y) \right) \varphi(y)$$

Since, the first product in the integrand clearly vanishes, the integral itself vanishes. Hence,

$$\frac{\partial u}{\partial x}(0,t) = 0$$

Thus, the restriction of u on $\mathbb{R}^+ \times \mathbb{R}$ satisfies the Neumann problem.

Finally, uniqueness follows by the energy method.

□

Example: We want to solve the Neumann problem for the initial data $\varphi = 1$. Whilst we can refer to the solution, an easier way to is recognize that $u(x,t) = 1$ solves the problem. Then, by uniqueness, $u(x,t) = 1$ is the solution.

□

Exercise 2.1.1. Solve $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}, u(x,0) = e^{-x}, u(0,t) = 0$ on the half life $0 < x < \infty$

Solution,

Here, we're given the problem:

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \quad u(x,0) = e^{-x} \quad u(0,t) = 0$$

on the half line $x \in (0, \infty)$.

For this problem, the general solution is:

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right] (e^{-x})$$

and we're done.

□

Exercise 2.1.2. Solve $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}, u(x,0) = 0, u(0,t) = 1$ for the half line $x \in (0, \infty)$

Solution,

We first use a change variable,

$$v(x,t) = u(x,t) - 1$$

Under this new variable, the problem reverts to the case where we solve using thhe odd extension. Now,

$$v(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right] (-1)$$

Thus,

$$u(x,t) = 1 - \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \mathrm{d}y \left[e^{-\frac{(x-y)^2}{4kt}} - e^{-\frac{(x+y)^2}{4kt}} \right]$$

□

Exercise 2.1.3. Consider the following problem with a Robin boundary condition:

$$\begin{aligned} \frac{\partial u}{\partial t} &= k\frac{\partial^2 u}{\partial x^2} \quad (x,t) \in (0,\infty) \times (0,\infty) \\ u(x,0) &= x \quad \frac{\partial u}{\partial x}(0,t) - 2u(0,t) = 0 \end{aligned}$$

The purpose of this exercise is to verify the solution formula for the problem. Let $f(x) = x$ for $x > 0$, and let $f(x) = x + 1 - e^{2x}$ for $x < 0$. Let,

$$v(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^\infty \mathrm{d}y \, e^{-\frac{(x-y)^2}{4kt}} f(y)$$

1. What PDE and initial condition does $v(x,t)$ satisfy for $x \in \mathbb{R}$.
2. Let $w = \frac{\partial v}{\partial x} - 2v$. What PDE and initial condition does $v(x,t)$ satisfy for $x \in \mathbb{R}^+$.

3. Show that $f'(x) - 2f(x)$ is an odd function (for $x \neq 0$)
4. Show that w is an odd function of x.
5. Deduce that $v(x,t)$ satisfies the problem. Assuming uniqueness, deduce the solution is given by:

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \mathrm{d}y \, e^{-\frac{(x-y)^2}{4kt}} f(y)$$

Solution,

(1.) Here,

$$f(x) = \begin{cases} x & x > 0 \\ x + 1 - e^{2x} & x < 0 \end{cases}$$

and define,

$$v(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \mathrm{d}y \, e^{-\frac{(x-y)^2}{4kt}} f(y)$$

Recall that the heat kernel,

$$\Phi(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-y)^2}{4kt}}$$

solves the heat equation:

$$\frac{\partial}{\partial t}\Phi = k\frac{\partial^2 \Phi}{\partial x^2}$$

So, differentiating under the integral sign:

$$\frac{\partial v}{\partial t} = \int_{-\infty}^{\infty} \mathrm{d}y \, \frac{\partial \Phi}{\partial t} f(y) = \int_{-\infty}^{\infty} \mathrm{d}y \, k\frac{\partial^2 \Phi}{\partial x^2} f(y) = k\frac{\partial^2}{\partial x^2} \int_{-\infty}^{\infty} \mathrm{d}y \, \Phi(x-y,t) f(y)$$

Therefore, we get:

$$\frac{\partial v}{\partial t} = k\frac{\partial^2 v}{\partial x^2}$$

The initial condition satisifed by v is:

$$v(x,0) = \lim_{t \to 0^+} v(x,t)$$

The heat kernel satisfies the concentration property so,

$$\lim_{t \to 0^+} \int_{\mathbb{R}} \mathrm{d}y \, \Phi(x-y,t) f(y) = f(x)$$